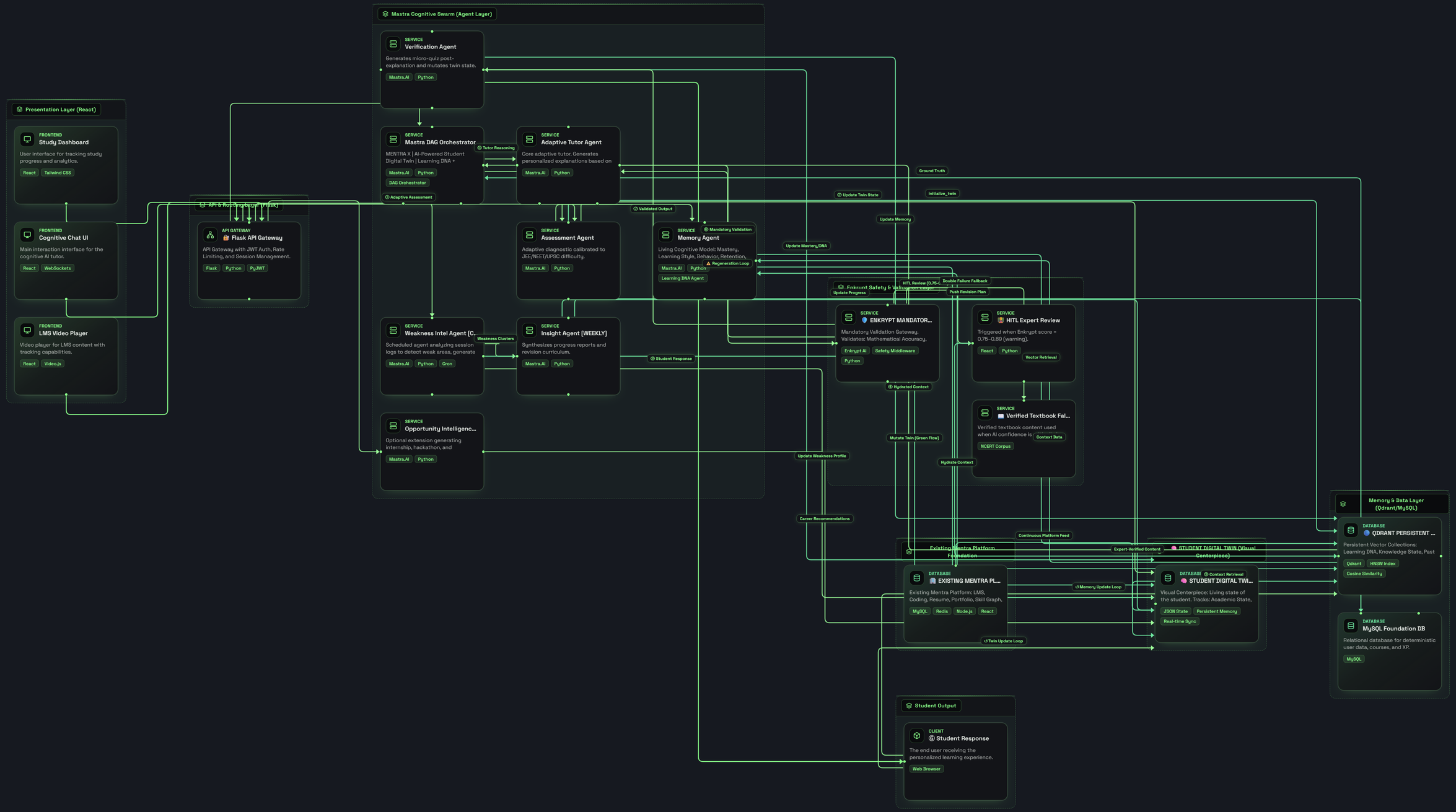
Mastra Cognitive Swarm (Agent Layer)
SERVICE
Verification Agent
Generates micro-quiz post-explanation and mutates twin state.
Mastra.AI
Python
Presentation Layer (React)
FRONTEND
Study Dashboard
User interface for tracking study progress and analytics.
React
Tailwind CSS
SERVICE
Mastra DAG Orchestrator
MENTRA X | AI-Powered Student Digital Twin | Learning DNA +
Mastra.AI
Python
DAG Orchestrator
Adaptive Assessment
Tutor Reasoning
SERVICE
Adaptive Tutor Agent
Core adaptive tutor. Generates personalized explanations based on
Mastra.AI
Python
Ground Truth
Update Twin State
initialize_twin
API & Routing Layer (Flask)
FRONTEND
Cognitive Chat UI
Main interaction interface for the cognitive AI tutor.
React
WebSockets
API GATEWAY
Flask API Gateway
API Gateway with JWT Auth, Rate Limiting, and Session Management.
Flask
Python
PyJWT
Validated Output
SERVICE
Assessment Agent
Adaptive diagnostic calibrated to JEE/NEET/UPSC difficulty.
Mastra.AI
Python
SERVICE
Memory Agent
Living Cognitive Model: Mastery, Learning Style, Behavior, Retention.
Mastra.AI
Python
Learning DNA Agent
Mandatory Validation
Regeneration Loop
Update Memory
Update Mastery/DNA
Enkrypt Safety & Validation Layer
HITL Review (0.75-0.89)
Double Failure Fallback
Update Progress
Push Revision Plan
FRONTEND
LMS Video Player
Video player for LMS content with tracking capabilities.
React
Video.js
SERVICE
Weakness Intel Agent [C...
Scheduled agent analyzing session logs to detect weak areas, generate
Mastra.AI
Python
Cron
Weakness Clusters
SERVICE
Insight Agent [WEEKLY]
Synthesizes progress reports and revision curriculum.
Mastra.AI
Python
Student Response
SERVICE
ENKRYPT MANDATOR...
Mandatory Validation Gateway. Validates: Mathematical Accuracy,
Enkrypt AI
Safety Middleware
Python
SERVICE
HITL Expert Review
Triggered when Enkrypt score = 0.75–0.89 (warning).
React
Python
Vector Retrieval
Hydrated Context
SERVICE
Verified Textbook Fal...
Verified textbook content used when AI confidence is
Context Data
NCERT Corpus
Mutate Twin (Green Flow)
Hydrate Context
SERVICE
Opportunity Intelligenc...
Optional extension generating internship, hackathon, and
Mastra.AI
Python
Update Weakness Profile
Career Recommendations
Memory & Data Layer (Qdrant/MySQL)
DATABASE
QDRANT PERSISTENT ...
Persistent Vector Collections: Learning DNA, Knowledge State, Past
Qdrant
HNSW Index
Cosine Similarity
Continuous Platform Feed
Existing Mentra Platform Foundation
Expert-Verified Content
STUDENT DIGITAL TWIN (Visual Centerpiece)
DATABASE
EXISTING MENTRA PL...
Existing Mentra Platform: LMS, Coding, Resume, Portfolio, Skill Graph,
MySQL
Redis
Node.js
React
Memory Update Loop
DATABASE
Context Retrieval
STUDENT DIGITAL TWI...
Visual Centerpiece: Living state of the student. Tracks: Academic State,
JSON State
Persistent Memory
Real-time Sync
Twin Update Loop
DATABASE
MySQL Foundation DB
Relational database for deterministic user data, courses, and XP.
MySQL
Student Output
CLIENT
Student Response
The end user receiving the personalized learning experience.
Web Browser